MSG No. 73.552 $Ib'c'a'$ [Type III, orthorhombic]

Table 1: Wyckoff site: `8a`, site symmetry: `-1'`

No.	position	mapping
1	$[0, 0, 0]$	`[1,5]`
2	$[0, 0, \frac{1}{2}]$	`[2,6]`
3	$[\frac{1}{2}, 0, 0]$	`[3,7]`
4	$[0, \frac{1}{2}, 0]$	`[4,8]`
5	$[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$	`[9,13]`
6	$[\frac{1}{2}, \frac{1}{2}, 0]$	`[10,14]`
7	$[0, \frac{1}{2}, \frac{1}{2}]$	`[11,15]`
8	$[\frac{1}{2}, 0, \frac{1}{2}]$	`[12,16]`

Table 2: Wyckoff site: `8b`, site symmetry: `-1'`

No.	position	mapping
1	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	`[1,13]`
2	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[2,14]`
3	$[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}]$	`[3,15]`
4	$[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}]$	`[4,16]`
5	$[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}]$	`[5,9]`
6	$[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}]$	`[6,10]`
7	$[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}]$	`[7,11]`
8	$[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}]$	`[8,12]`

Table 3: Wyckoff site: `8c`, site symmetry: `2..`

No.	position	mapping
1	$[x, 0, \frac{1}{4}]$	`[1,2]`
2	$[\frac{1}{2} - x, 0, \frac{3}{4}]$	`[3,12]`
3	$[-x, \frac{1}{2}, \frac{1}{4}]$	`[4,11]`
4	$[-x, 0, \frac{3}{4}]$	`[5,6]`
5	$[x + \frac{1}{2}, 0, \frac{1}{4}]$	`[7,16]`
6	$[x, \frac{1}{2}, \frac{3}{4}]$	`[8,15]`
7	$[x + \frac{1}{2}, \frac{1}{2}, \frac{3}{4}]$	`[9,10]`
8	$[\frac{1}{2} - x, \frac{1}{2}, \frac{1}{4}]$	`[13,14]`

Table 4: Wyckoff site: `8d`, site symmetry: `.2.`

No.	position	mapping
1	$[\frac{1}{4}, y, 0]$	`[1,3]`
2	$[\frac{1}{4}, -y, \frac{1}{2}]$	`[2,12]`
3	$[\frac{3}{4}, \frac{1}{2} - y, 0]$	`[4,10]`
4	$[\frac{3}{4}, -y, 0]$	`[5,7]`
5	$[\frac{3}{4}, y, \frac{1}{2}]$	`[6,16]`

continued ...

Table 4

No.	position	mapping
6	$\left[\frac{1}{4},\, y+\frac{1}{2},\, 0\right]$	`[8,14]`
7	$\left[\frac{3}{4},\, y+\frac{1}{2},\, \frac{1}{2}\right]$	`[9,11]`
8	$\left[\frac{1}{4},\, \frac{1}{2}-y,\, \frac{1}{2}\right]$	`[13,15]`

Table 5: Wyckoff site: `8e`, site symmetry: `..2`

No.	position	mapping
1	$\left[0,\, \frac{1}{4},\, z\right]$	`[1,4]`
2	$\left[0,\, \frac{3}{4},\, \frac{1}{2}-z\right]$	`[2,11]`
3	$\left[\frac{1}{2},\, \frac{1}{4},\, -z\right]$	`[3,10]`
4	$\left[0,\, \frac{3}{4},\, -z\right]$	`[5,8]`
5	$\left[0,\, \frac{1}{4},\, z+\frac{1}{2}\right]$	`[6,15]`
6	$\left[\frac{1}{2},\, \frac{3}{4},\, z\right]$	`[7,14]`
7	$\left[\frac{1}{2},\, \frac{3}{4},\, z+\frac{1}{2}\right]$	`[9,12]`
8	$\left[\frac{1}{2},\, \frac{1}{4},\, \frac{1}{2}-z\right]$	`[13,16]`

Table 6: Wyckoff site: `16f`, site symmetry: `1`

No.	position	mapping
1	$[x,\, y,\, z]$	`[1]`
2	$\left[x,\, -y,\, \frac{1}{2}-z\right]$	`[2]`
3	$\left[\frac{1}{2}-x,\, y,\, -z\right]$	`[3]`
4	$\left[-x,\, \frac{1}{2}-y,\, z\right]$	`[4]`
5	$[-x,\, -y,\, -z]$	`[5]`
6	$\left[-x,\, y,\, z+\frac{1}{2}\right]$	`[6]`
7	$\left[x+\frac{1}{2},\, -y,\, z\right]$	`[7]`
8	$\left[x,\, y+\frac{1}{2},\, -z\right]$	`[8]`
9	$\left[x+\frac{1}{2},\, y+\frac{1}{2},\, z+\frac{1}{2}\right]$	`[9]`
10	$\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, -z\right]$	`[10]`
11	$\left[-x,\, y+\frac{1}{2},\, \frac{1}{2}-z\right]$	`[11]`
12	$\left[\frac{1}{2}-x,\, -y,\, z+\frac{1}{2}\right]$	`[12]`
13	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, \frac{1}{2}-z\right]$	`[13]`
14	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, z\right]$	`[14]`
15	$\left[x,\, \frac{1}{2}-y,\, z+\frac{1}{2}\right]$	`[15]`
16	$\left[x+\frac{1}{2},\, y,\, \frac{1}{2}-z\right]$	`[16]`